Product & Science: Senolytic Regenerative Therapy (SRT)

What the Product Is

A Senolytic Regenerative Therapy (SRT) is a pharmaceutical treatment course that combines two mechanisms:

• Senolytic clearance: Small molecules that selectively induce apoptosis

(programmed cell death) in senescent cells -- damaged, non-dividing cells that accumulate with age and secrete inflammatory signals (the "senescence-associated secretory phenotype," or SASP).

• Telomere stabilization: Peptide compounds that activate telomerase in stem cell

compartments, slowing or partially reversing the shortening of telomere caps on chromosomes -- a primary driver of cellular aging.

Together, these produce measurable reversal of biological aging: organs function as though the patient were years younger, chronic inflammation drops, tissue regeneration improves, and age-related disease risk declines.

How It Is Administered

Gen 1 (2031 launch)

• Route: Intravenous infusion (IV) administered at a certified clinic.

• Regimen: One infusion per quarter (4 per year), each taking 4–6 hours.

• Monitoring: Patients require 48-hour post-infusion observation for acute

reactions. Blood panels at weeks 2, 6, and 12 post-infusion.

• Cold chain: The active compound (lyophilized) must be stored at -20°C and

reconstituted on-site within 4 hours of thawing.

This delivery method is expensive and inconvenient, limiting the patient base to those who can afford both the drug and the clinic infrastructure.

Future Generations (R&D dependent)

Generation	Route	Regimen	Monitoring
Gen 2	Subcutaneous injection (s	Monthly	Quarterly blood panel
Gen 3	Oral tablet (daily)	Daily pill	Semi-annual blood panel
Gen 4	One-time gene therapy + a	Single procedure + daily	Annual check-up

Each generation advance requires substantial R&D investment and regulatory approval.

Efficacy by Generation

Efficacy is measured primarily by epigenetic age reversal (Horvath clock, years) and functional biomarkers (grip strength, VO2max, cognitive scores).

Metric	Gen 1	Gen 2	Gen 3	Gen 4
Epigenetic age rever	5–8	10–15	15–20	20–25
VO2max improvement	20–30%	35–50%	50–70%	70–90%

Cognitive score impr	10–15%	20–30%	30–45%	45–60%
Onset of benefit	3–6 months	1–3 months	2–4 weeks	1–2 weeks
Duration if treatmen	Reversal over 18–24	Reversal over 24–36	Partial permanence	Largely permanent

Important: These are population averages. Individual response varies by ±30% depending on baseline health, genetics, and lifestyle. Some patients are "super-responders" (2x average benefit); roughly 5% are "non-responders."

Side-Effect Profile

Side effects are the critical barrier to mass adoption. They fall into categories:

Acute (within 72 hours of infusion/dose)

Side Effect	Gen 1 Rate	Gen 2 Rate	Gen 3 Rate	Gen 4 Rate
Fatigue and malaise	45%	25%	10%	5%
Nausea/vomiting	20%	8%	3%	1%
Fever (immune activa	15%	5%	2%	<1%
Injection site react	N/A (IV)	10%	N/A (oral)	N/A

Serious (Grade 3+)

Side Effect	Gen 1 Rate	Gen 2	Gen 3	Gen 4
Peripheral neuropath	3.1%	1.0%	0.2%	<0.1%
Autoimmune flare (im	2.4%	0.8%	0.15%	<0.05%
Severe fatigue (hosp	1.8%	0.5%	0.1%	<0.05%
Partial paralysis (t	0.4%	0.05%	<0.01%	<0.01%
Organ toxicity (live	0.3%	0.1%	0.02%	<0.01%
Total serious AE rat	7.3%	2.5%	<0.5%	<0.2%
Death	0.02% (1 in 5,000)	<0.005%	<0.001%	<0.001%

The Paralysis Problem

The most feared side effect is transient motor neuropathy -- partial paralysis of one or more limbs, typically resolving in 2–6 months but occasionally persisting longer. The mechanism: senolytic compounds can inadvertently trigger apoptosis in Schwann cells (which form the myelin sheath around peripheral nerves) when they are in a "quasi-senescent" repair state.

This side effect is:

- Rare (0.4% in Gen 1) but terrifying to patients and the public.
- Unpredictable: No reliable biomarker identifies who is at risk.
- Mediagenic: Every case generates news coverage and social media virality.
- The primary R&D target: Reducing this side effect is the #1 priority for Gen 2

development. Firms that solve it first gain enormous competitive advantage.

Side-Effect Impact on Demand

Consumer willingness to pay is heavily influenced by the serious AE rate:

Serious AE Rate	Demand Modifier	Market Perception
>5%	0.5x baseline	"Experimental, risky"
3–5%	0.7x baseline	"Promising but dangerous"

1–3%	1.0x baseline	"Acceptable for high-value therapy
0.5–1%	1.5x baseline	"Safe enough for broader adoption"
<0.5%	2.5x baseline	"Routine medical procedure"
<0.1%	4.0x baseline	"Over-the-counter safe"

Product Quality Dimensions

From a consumer's perspective, product quality has three components that firms can invest in:

1. Efficacy (how well it works)

Driven by R&D investment in next-generation compounds. Higher efficacy = more age reversal = higher willingness to pay. This is the dimension that unlocks premium pricing.

2. Safety (how few side effects)

Driven by R&D investment in targeted delivery and compound refinement. Lower serious AE rate = larger addressable market + higher willingness to pay. This is the dimension that unlocks volume.

3. Convenience (how easy to take)

Driven by R&D in formulation and delivery. IV infusion -> subcutaneous -> oral -> one-time. Lower convenience = higher patient burden = lower adherence = lower effective demand.

A firm's overall product quality score (used in the demand system) is a composite of these three dimensions, weighted by what the market values:

```
quality_score = w_efficacy * efficacy_index
              + w_safety * safety_index
              + w_convenience * convenience_index
```

Weights shift over time: early market values efficacy most (desperate patients will tolerate side effects); mature market values safety and convenience most.

Intellectual Property

• Each firm's specific compound formulation is patented (20-year patent life).

• The underlying scientific mechanism (senolytic clearance + telomere stabilization) is
not patentable -- it is published academic knowledge.

• Firms compete on proprietary formulations, manufacturing processes, and
delivery systems, not on the basic science.

• Patent expiration is not modeled explicitly in the 20-year simulation, but firms
should be aware that long-term competitive advantage comes from continuous innovation, not from resting on initial IP.

What Firms Need to Know for Decision-Making

• Your Gen 1 product is mediocre by future standards. It works, but it is
expensive to make, has significant side effects, and requires clinic visits. Standing still means being overtaken.

• R&D is the path to competitive advantage, but it is expensive and uncertain.

Not every R&D program succeeds. You must balance current profitability with long-term investment.

- Side effects are the #1 demand driver. A firm that achieves Gen 2 safety (2.5% serious AE) while competitors are at Gen 1 (7.3%) will see a large demand shift in its favor.
- The market grows as the product improves. You are not fighting over a fixed pie -- you are expanding the pie by making the product better and cheaper.
- Public perception matters. A paralysis case at your firm specifically (not just industry-wide) can trigger a demand shock for your product. Marketing and physician education can partially buffer this.